

ECA5601 –Advanced Wireless Communication Systems / 5G / 6G

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Experiment: 9

Massive MIMO Capacity vs SNR and Gain Using MATLAB

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Aim:To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objectives: Analyze Channel Capacity (bps/Hz)

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

Software Required: • MATLAB

Theory:

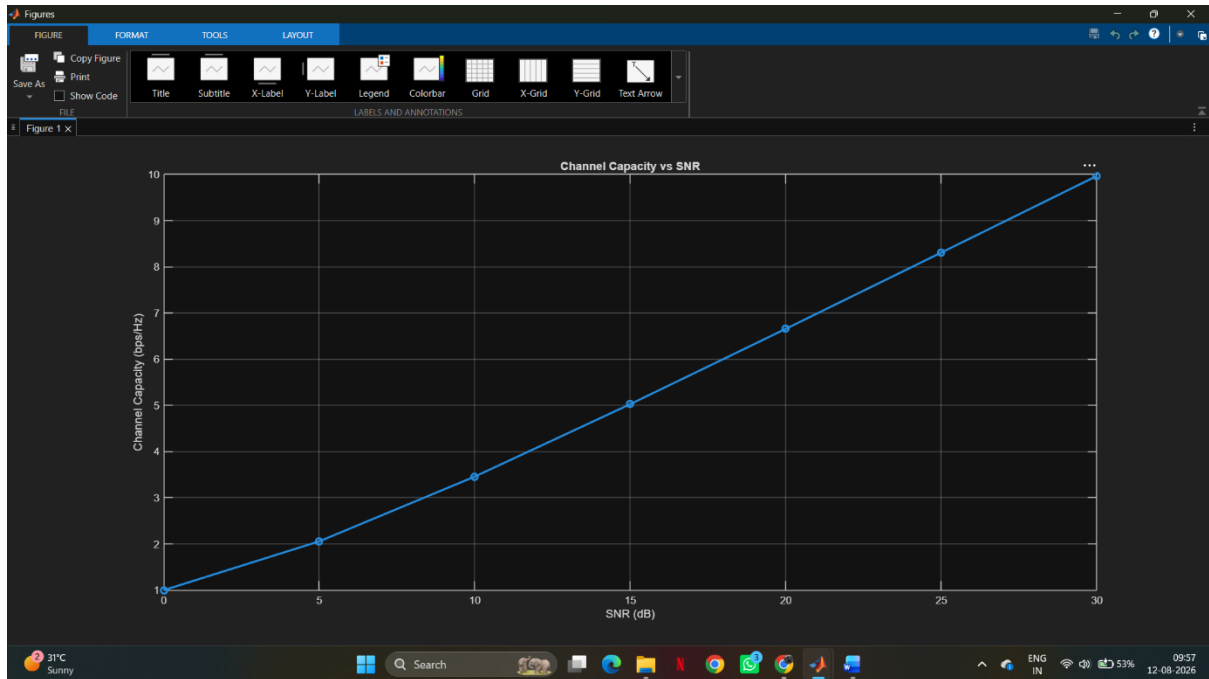
Massive MIMO employs a large number of antennas at the base station to simultaneously serve multiple users. The increased spatial diversity and beamforming capability significantly improve channel capacity, signal quality, and antenna gain. Channel capacity is influenced by the signal-to-noise ratio (SNR), while antenna gain increases with the number of antenna elements. These parameters are fundamental to achieving the high throughput and spectral efficiency required in 5G and beyond wireless communication systems.

MATLAB Code:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
```

```
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);  
  
grid on;  
  
xlabel('SNR (dB)');  
  
ylabel('Channel Capacity (bps/Hz)');  
  
title('Channel Capacity vs SNR');
```

Output:



Result:

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.